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Barriers to Gender Convergence*

The Interactive Effects of Job Inflexibility and Social Norms

Kazuharu Yanagimoto 

Kobe University

yanagimoto@econ.kobe-u.ac.jp

ABSTRACT This paper investigates the barriers to gender convergence using Japan as a salient environment to explore the interactive effects of labor market structures and social norms. I develop a quantitative model of household labor supply where couples jointly decide their occupations and working hours. The model features a labor market with inflexible “regular” jobs with convex pay schedules and flexible “non-regular” jobs, interacting with social norms regarding spousal earnings. The calibrated model successfully reproduces observed gender gaps in participation, occupation, and working hours, and explains 56% of the gender wage gap. The model also accounts for cross-regional differences in gender gaps solely through variation in social norms. Counterfactual simulations show that while increasing job flexibility substantially reduces wage and occupational gaps, the working hours gap persists due to the unequal burden of domestic work. Closing this remaining gap requires policies such as affordable household services. Furthermore, the model predicts that structural reforms are particularly effective in societies with strong gender norms.

Keywords: Gender Wage Gap, Social Norms, Job Inflexibility, Home Production

JEL Codes: J16, J22, J31

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1 Introduction

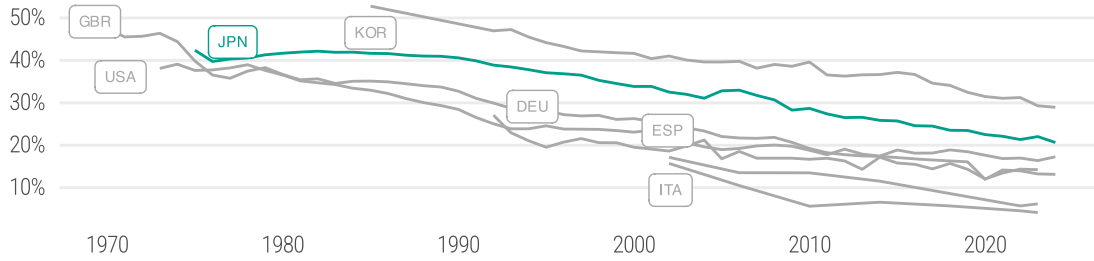
Despite significant progress in gender equality over the past century, gender gaps in labor market outcomes persist in many developed economies. Recent literature suggests that these remaining disparities are correlated with social norms regarding gender roles (Kleven, Landais, and Sogaard 2019; Kleven et al. 2019; Kleven, Landais, and Leite-Mariante 2025). While this correlation is well-documented, the precise mechanisms through which social norms interact with labor market institutions to generate these gaps remain less understood.

In this paper, I argue that Japan provides an ideal setting to explore these interactive effects. Japan is characterized by particularly strong gender norms and a distinct labor market structure. Unlike other high-income countries where gender gaps have narrowed, Japan's gender gap remains persistently high, suggesting that the forces hindering convergence are particularly salient here. As I document in Section 2.4, social norms regarding gender roles are pronounced in Japan. Furthermore, the Japanese labor market is characterized by a strict distinction between *regular* and *non-regular* employment, which creates a sharp trade-off between wages and flexibility.

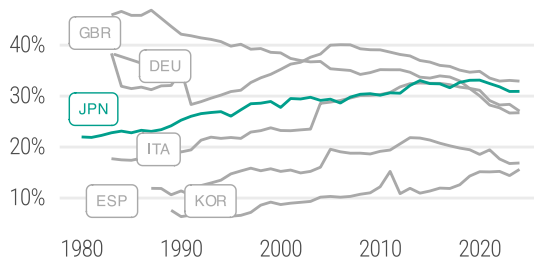
Figure 1 illustrates the first point: the persistence of gender gaps. Panel (a) shows the gender gap in median earnings of full-time employees across several high-income countries. While these gaps are gradually closing worldwide, Japan shows little convergence with international trends. The earnings gender gap in Japan remained at 21.3% in 2022, substantially higher than most high-income countries (with Korea being the only exception).

Simultaneously, as shown in panel (b), the share of women in part-time employment has steadily increased in Japan, reaching 31.8% in 2022, higher than in other high-income countries. Interestingly, panel (c) reveals that Japan maintains one of the highest female employment rates among high-income countries, at 81.8% in 2022. Thus, high female labor force participation in Japan goes together with high part-time work and significant gender gaps in earnings (Teruyama, Goto, and Lechevalier 2018). In the Japanese context, this “part-time” work is largely synonymous with non-regular employment, a category defined by contract status rather than just hours, which relates to the third point about the labor market structure.

(a) Gap in Median Earnings of Full-time Workers



(b) Part-time Employment



(c) Employment Rate of Women

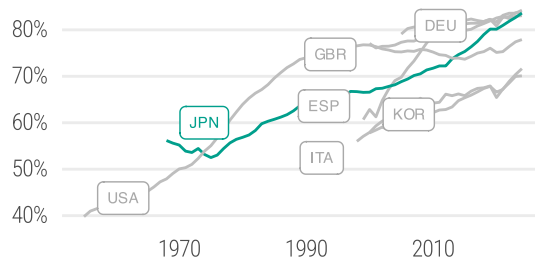


Figure 1: Labor Market in Japan. (a) Gender gap in median earnings of full-time employees relative to male earnings. (b) Share of part-time employment in total employment. Part-time employment is defined as usually working less than 30 hours per week in the main job. (c) Employment rate, calculated as the ratio of employed persons to the working-age population. Sample restricted to ages 25-54. Source: OECD.

To explain why Japanese women disproportionately sort into low-paying, non-regular employment despite high participation rates, I first highlight the institutional distinction between **regular** and **non-regular** employment. Unlike the full-time/part-time distinction common in other economies, which is often based on hours worked, the regular/non-regular dichotomy in Japan is defined by contract type. Regular jobs typically offer permanent contracts and higher wages but demand long working hours and inflexible schedules. In contrast, non-regular jobs are characterized by fixed-term contracts and lower wages but provide significantly greater flexibility in terms of hours and location. I document that this institutional structure creates a stark trade-off between earnings and flexibility.

Second, I examine the role of social norms, specifically the male breadwinner norm. I document a sharp discontinuity in the distribution of wives' earnings relative to their husbands' at the 50% threshold, suggesting a strong aversion to wives outearning their husbands. This "gender identity" (Bertrand, Kamenica, and Pan 2015) reinforces the traditional division of labor, where women assume the primary responsibility for domestic work.

I argue that structural barriers (job inflexibility) and cultural barriers (social norms) are mutually reinforcing. The convexity of regular wage schedules makes short hours costly, while social norms make long hours for women socially penalized. This interaction creates a "missing middle" in the choice set for women: they must choose between a high-commitment regular job that violates norms or a low-commitment non-regular job that entails a significant wage penalty. Women,

constrained by social norms to prioritize household responsibilities or limit their earnings, are effectively priced out of regular employment and pushed into non-regular jobs, which offer the necessary flexibility but at a steep wage penalty.

To quantify these mechanisms, I develop a model of household labor supply where couples jointly decide their occupations, working hours, and domestic labor hours. In this framework, each partner is endowed with a productivity level, and the household faces a specific domestic labor requirement. Couples choose between regular employment, non-regular employment, or non-employment. Regular jobs feature a convex wage schedule that rewards long hours, while non-regular jobs offer linear compensation but are subject to a productivity penalty. Critically, it incorporates a utility cost when a wife's earnings exceed her husband's, capturing the influence of social norms.

In this economy, couples weigh the benefits of higher potential earnings in regular jobs against the costs of inflexible hours and social norm penalties. The model generates realistic patterns of sorting, where husbands typically hold regular jobs while wives often select non-regular positions or exit the labor market to fulfill domestic responsibilities. The model's parameters are calibrated to match observed correlations between husbands' and wives' wages and working hours, as well as the joint distribution of couples' earnings.

After calibration, I evaluate the model's performance by comparing its predictions against empirical gender gaps, outcomes that were deliberately *not targeted* during the calibration process. The baseline model successfully explains a substantial proportion of observed gender disparities: nearly all gaps in participation rates, occupational choices, and labor hours, along with 55.6% of the wage gap. Additionally, the model accurately reproduces both the joint distribution of couples' occupational choices (across regular, non-regular, and non-working categories) and the joint distribution of working hours conditional on these occupational selections. Furthermore, I exploit the regional variation in social norms across Japanese prefectures to validate the model's mechanism. I show that the model, by varying only the social norm parameter, can account for a significant portion of the cross-regional variation in gender gaps.

What role do job inflexibility and social norms play in perpetuating these gender gaps, and how can these disparities be mitigated? To address these questions, I conduct two counterfactual simulations. First, I simulate a scenario with **flexible regular jobs**, where the wage schedule is linear rather than convex. I find that this structural change substantially reduces gender gaps in participation, occupation, and wages. However, the gender gap in working hours remains, as social norms continue to impose a disproportionate domestic burden on women. Second, I examine the impact of **outsourcing housework**. Japanese couples rarely utilize external housework services and consequently devote substantial time to domestic tasks, particularly when raising young children. One contributing factor to this pattern is Japan's highly restricted international migration, which limits the availability of household labor.² By allowing households to purchase affordable household services, this policy eliminates significant proportions of all gender gaps, including the working hours gap.

²Several studies examine the impact of low-skilled migration on women's labor supply, including Furtado (2016); Cortés and Tessada (2011); Cortés and Pan (2019).

Finally, I investigate the interaction between structural reforms and social norms. I find that the effectiveness of these structural changes, such as increasing job flexibility or subsidizing outsourcing, is heterogeneous depending on the strength of social norms. Specifically, these policies are more effective in reducing gender gaps when social norms are stronger. This implies that structural reforms are particularly crucial for societies with conservative gender norms, such as Japan, as they can mitigate the adverse effects of cultural barriers.

Related Literature This paper contributes to the growing body of labor and macroeconomic literature examining the relationship between household responsibilities and gender gaps in the labor market.³

First, this paper relates to the literature linking social norms to gender gaps. Kleven, Landais, and Sogaard (2019) and Kleven et al. (2019) document that a large portion of the remaining gender inequality is due to “child penalties,” and that these penalties are correlated with gender norms. Bertrand, Kamenica, and Pan (2015) argue that gender identity norms, specifically the aversion to wives earning more than their husbands, are a key driver of gender gaps in the US. I contribute to this literature by explicitly modeling how these norms interact with labor market structures. Specifically, I demonstrate that social norms regarding gender roles compel women to assume a disproportionate share of domestic work, while job inflexibility further exacerbates these disparities. By elucidating these interactions, I provide insights into the mechanisms driving marriage and child penalties, highlighting how social norms and flexibility in occupations reinforce each other to generate gender disparities.

Second, it connects to the literature on job flexibility and non-linear wages. Goldin and Katz (2011) and Goldin (2014) identify that certain occupations feature convex (non-linear) wage schedules while others maintain linear relationships between hours and compensation. This distinction aligns with empirical evidence showing that part-time workers typically earn less per hour than their full-time counterparts (Aaronson and French 2004; Ameriks et al. 2020). Building on these insights, Erosa et al. (2022) models couples’ occupational decisions across jobs with varying degrees of flexibility. While their framework effectively captures heterogeneous job flexibility through a streamlined approach, it treats the allocation of home hours as exogenous, assuming women inherently dedicate more time to household tasks than men. In contrast, my model endogenously generates these differences through its home production component. Another relevant contribution comes from Cubas, Juhn, and Silos (2023), who model the concentration of working schedules by incorporating penalties for absence during peak hours. They argue that women with children face disproportionate penalties due to their greater household responsibilities. While their approach models job flexibility using detailed time-use data, my framework conceptualizes flexibility through the convexity of wage schedules.

Finally, this paper significantly advances our economic understanding of gender gaps in Japan. Despite having one of the largest gender disparities among developed nations, the underlying drivers of Japan’s gender inequality have received limited attention in economic literature. Onozuka (2016) examines the partial convergence in Japan’s gender wage gap from 1992 to 2002, arguing

³For a comprehensive recent review, see Albanesi, Olivetti, and Petrongolo (2023).

that women were systematically displaced from regular to non-regular employment during this period. Teruyama, Goto, and Lechevalier (2018) identifies increased female labor supply as a primary factor in the growth of non-regular employment in Japan during the 2000s. Additionally, Kitao and Mikoshiba (2022) provides quantitative analysis of how fiscal policies influence female labor force participation and occupational choices. While these studies offer partial explanations for gender disparities in occupation and wages, few have comprehensively disentangled the structural causes of Japan’s gender gaps, and most notably, they have not adequately addressed the role of social norms. To my knowledge, this research represents the first comprehensive analysis that simultaneously explains Japan’s gender gaps across four critical dimensions: labor force participation, occupational selection, working hours, and wages.

The remainder of this paper is structured as follows: In the next section, I document key empirical facts about gender gaps in Japan. In Section 3, I develop the baseline model that captures couples’ joint decisions on occupational choices and working hours while incorporating social norms. Section 4 outlines the calibration methodology. Section 5 presents the quantitative results of the baseline model and evaluates its performance in explaining the gender gaps. Section 6 validates the model mechanism using regional variation in social norms. In Section 7, I conduct counterfactual simulations to explore the mechanisms driving these gaps, specifically focusing on the roles of job inflexibility, social norms, and the outsourcing of housework. Section 8 summarizes the findings and discusses their implications.

2 Stylized Facts

This chapter presents descriptive evidence on the sources of gender gaps in the Japanese labor market. To this end, I introduce the distinction between **regular** and **non-regular** employment, a key institutional feature of the Japanese labor market. I document significant gender differences in the selection into these job types and show that this sorting reflects a fundamental trade-off between wage and flexibility. Furthermore, I highlight the role of social norms, particularly the male breadwinner norm, as a root cause of these gender disparities.

2.1 Data

This paper mainly relies on the Japanese Panel Study of Employment Dynamics (JPSED). The JPSED is a panel dataset of individual workers since 2016, with the most recent data wave from 2025.⁴ The sample consists of 57,284 individuals older than 15. This survey contains information on earnings, working hours, domestic labor hours, and types of jobs. For married individuals, it also contains information on the spouse’s job and earnings.

For the main analysis, I use the pooled data of JPSED2017-2020. The JPSED2016 does not have information on the domestic labor hours, which is important for this analysis. The JPSED2021 and later waves may be affected by the COVID-19 pandemic. The sample includes married men and women aged 25 to 59 who are in the labor market.

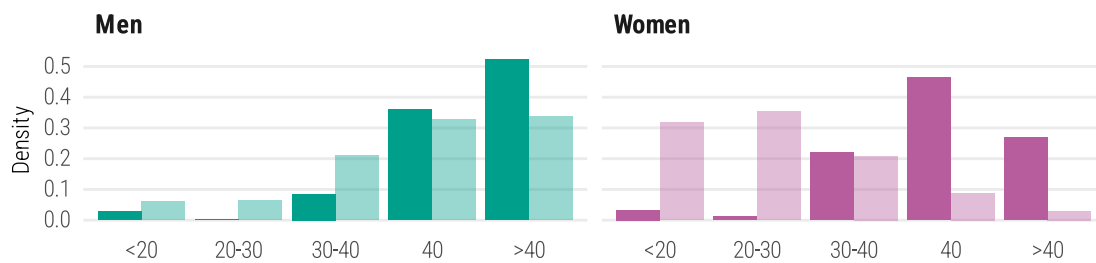
⁴Data were distributed one year later. The data in 2024 was distributed in 2025 and called JPSED2025.

2.2 Regular and Non-Regular Workers

In the context of Japanese statistics, the terms **regular** and **non-regular** jobs are widely used to categorize employment types. As this classification is based on each company's internal categorization, there are no precise legal definitions.⁵ However, they are typically described as follows: A regular worker usually has a permanent contract, works 40 hours or more at a higher wage, while a non-regular worker has a temporary contract and works less than 40 hours at a lower wage.

Panel (a) in Figure 2 shows that 36.1% (46.4%) of male (female) regular workers work exactly 40 hours per week and 52.3% (26.8%) work more than that. On the other hand, only 33.7% (3.0%) of non-regular male (female) workers work more than 40 hours. In addition, panel (b) shows that these two occupations differ in hourly wages. While 33.1% (66.6%) of non-regular male (female) workers earn a wage of less than 1000 JPY⁶, only 4.0% (17.0%) of regular workers do. Note that using the OECD definition of “less than 30 hours per week in their main job,” 16.5% (74.4%) of male (female) non-regular workers would be categorized as being in part-time employment.

Weekly Working Hours



Hourly Wage (JPY)

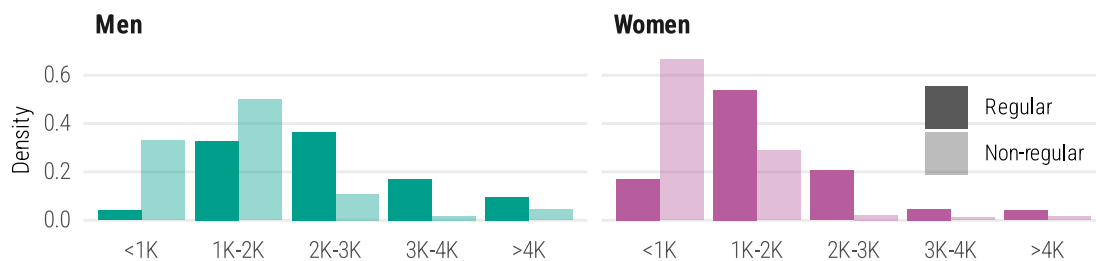


Figure 2: Distribution of Weekly Working Hours and Hourly Wage. The data is pooled from JPSED 2017-2020. The sample includes married men and women aged 25-59.

The gender gap in occupational choices is shown in Figure 3. We can see clear gender differences in occupational choices in married individuals. While almost 90% of men work as regular workers, less than 30% of women work as regular workers, and the share decreases with age. In addition, the proportion of non-regular workers is much higher among female employees; almost half of the female employees choose non-regular jobs.

⁵See Asao (2011) for a more detailed discussion on the definition of regular and non-regular jobs.

⁶1000 JPY $\approx$ 5.56 EUR.

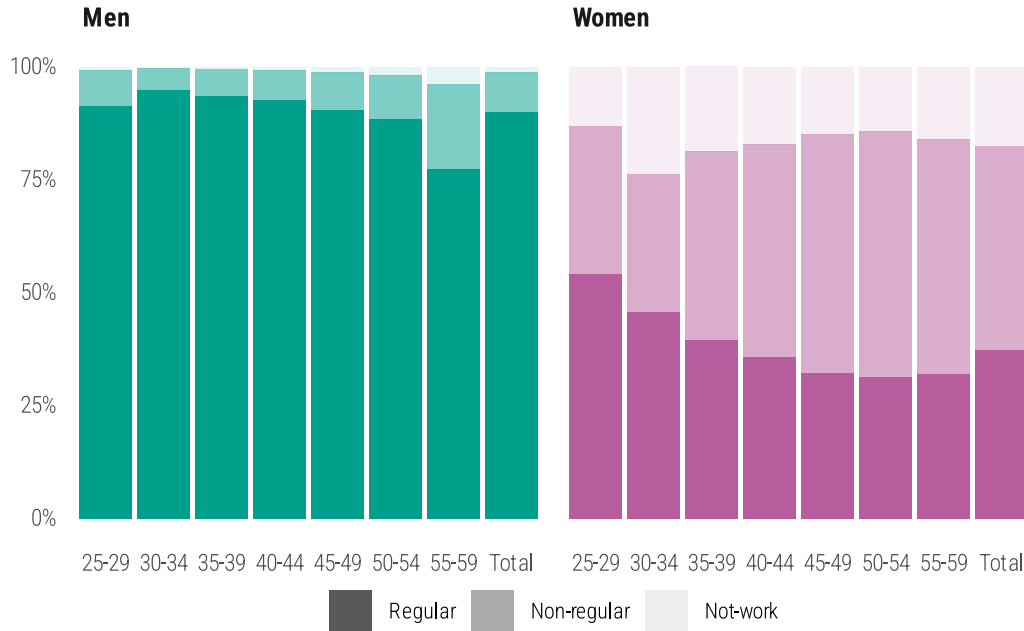


Figure 3: Occupational Choice of Married Individuals. The data is pooled from JPSED 2017-2020. The sample includes married men and women aged 25-59.

Given the substantial disparities in wages and working hours between regular and non-regular employment, this divergence in occupational choice serves as a primary driver of the gender wage gap. Furthermore, the occupational segregation itself represents a distinct and critical dimension of gender inequality in the labor market.

Differences from Full-time and Part-time Employment While economic literature often distinguishes between full-time and part-time employment, I focus on the distinction between regular and non-regular jobs. The full-time/part-time dichotomy is typically defined by hours worked (e.g., the OECD defines full-time as working 30 hours or more per week). If workers can freely choose their hours, this classification becomes an endogenous outcome of their labor supply decisions. In contrast, the regular/non-regular distinction is an exogenous categorization defined by the employer and employment contract, representing distinct choice sets available to workers. As discussed in Section 2.3, these two job types differ fundamentally in their wage structures and flexibility, presenting workers with a clear trade-off.

2.3 Job Flexibility

Why do women choose non-regular jobs? To answer this question, I map regular and non-regular jobs into non-linear and linear jobs in Goldin (2014), who emphasizes that some jobs have a highly non-linear (convex) pay structure with respect to working hours, while others have a nearly linear one. Given a non-linear wage schedule, a worker can work at a high wage in exchange for long working hours. On the other hand, a linear job worker can flexibly decide their working hours since there is no wage penalty for reducing working hours. Hence, there is a trade-off between job flexibility and wage.

Figure 4 represents a direct measurement of the flexibility of regular and non-regular jobs, based on survey data. The questions asked regular and non-regular workers about the flexibility of their jobs on a 5-point scale: 5 is the highest and 1 is the lowest. Each point shows the mean job flexibility score in terms of working days, working hours, and working place. We can see that the regular worker has less flexibility in all aspects. Also, female non-regular workers have more flexibility than male regular workers, while female regular workers face the same inflexibility as male regular workers.



Figure 4: Flexibility of Regular and Non-regular Jobs. Pooled data from JPSED 2017-2020. The sample includes married men and women aged 25-59. The figure plots the mean score of job flexibility (1: Inflexible to 5: Flexible) regarding working days, hours, and place.

Finally, if regular jobs require 40 hours of commitment and women have to allocate a large amount of time to housework, they may choose non-regular jobs, which are more flexible. Indeed, Figure 5 supports this argument. This figure shows the reasons why married women in the non-regular workforce chose their current job. More than 60% of the women chose “job flexibility” as the reason and nearly 40% chose “housework”. We can also see “cannot get regular jobs” is not the main reason (less than 10%) why they chose non-regular jobs.

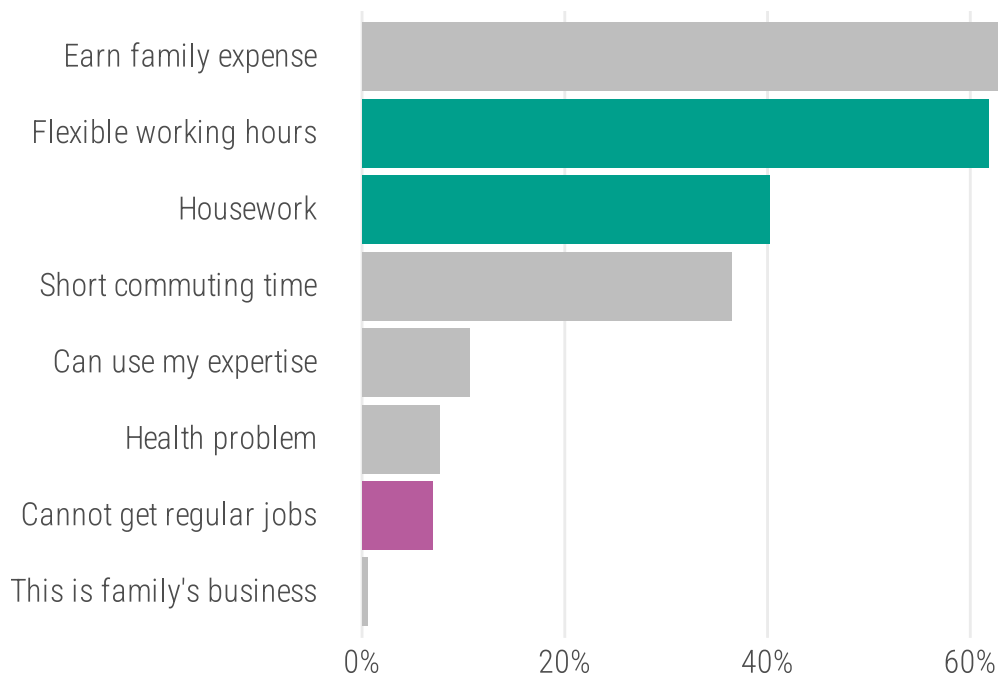


Figure 5: Reasons Why Women Choose Non-regular Work. The data is pooled from JPSED 2017-2020. The sample includes married women aged 25-59 who have non-regular jobs. Respondents could select multiple reasons.

While job inflexibility is not unique to Japan, it is more severe than in other countries. Since there is no survey asking the same question about job inflexibility in Japan and other countries, as a proxy measurement, I use a question in JPSED and the Labor Force Survey of Eurostat. In the JPSED, respondents are asked to rate their agreement with the statement “I was able to choose my working days” on a scale of 1 to 5. Eurostat includes a question on “Persons in employment by level of difficulty to take one or two hours off at short notice”, the respondents answer on a 1-4 scale. To match the JPSED sample to Eurostat, I use men and women employed, aged between 35 and 49, and having children, and rescale the level of difficulty to the 1-4 scale. Although this is not a direct comparison between Japan and other countries, Figure 6 indicates that regular workers in Japan might have more difficulty taking a day off.

Level of difficulty to take one or two days of leave at short notice.

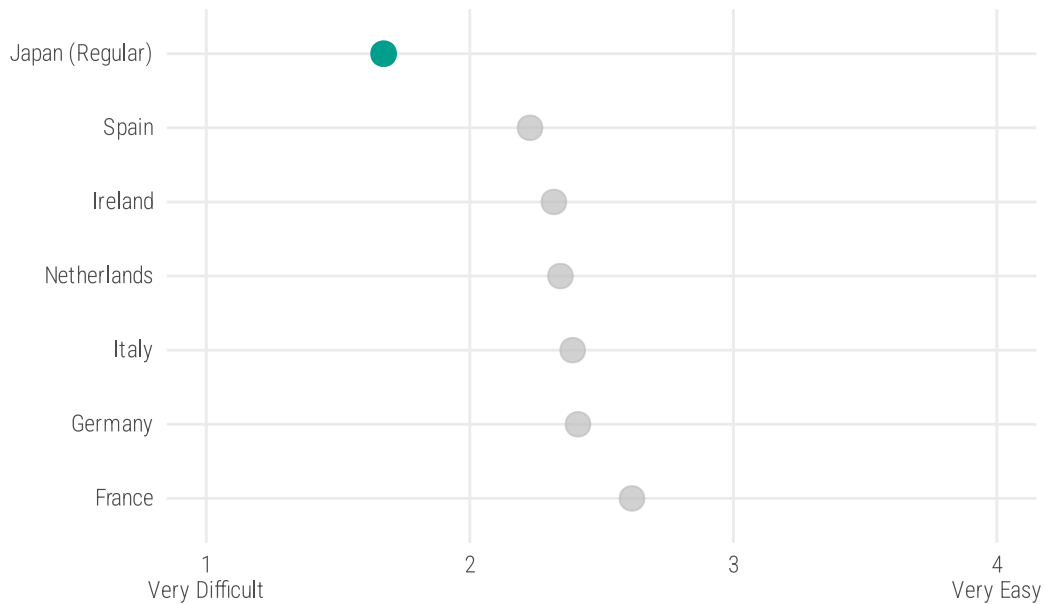


Figure 6: Flexibility of Working Days. The sample includes employed men and women aged 35-49 with children. Data sources are the Eurostat Labor Force Survey (2019) for European countries and JPSED for Japan. For Japan, the sample is restricted to regular workers.

2.4 Social Norms

While the difference in flexibility between occupations is crucial, these job characteristics should, in principle, be symmetric for men and women. From where, then, does the gender disparity originate? In this paper, I focus on social norms regarding gender roles, specifically the male *breadwinner* norm: the expectation that husbands should earn more than their wives. This can be interpreted as the flip side of the norm that women should bear the primary responsibility for housework and childcare.

Bertrand, Kamenica, and Pan (2015) show that there is a sharp drop in the distribution of household income share at the point where wives earn more than their husbands. This type of gender norm plays a similar role in Japan. Figure 7 shows the distribution of the earning share of wives. We observe a clear discontinuity between the distribution below and above 50%, which suggests discontinuous behavior at the point where a wife earns more than her husband.⁷ Furthermore, the bunching pattern just below 50% supports the strength of this social norm.

⁷I conduct robustness check with smaller bins. The results, which are similar, are in Section A.1.

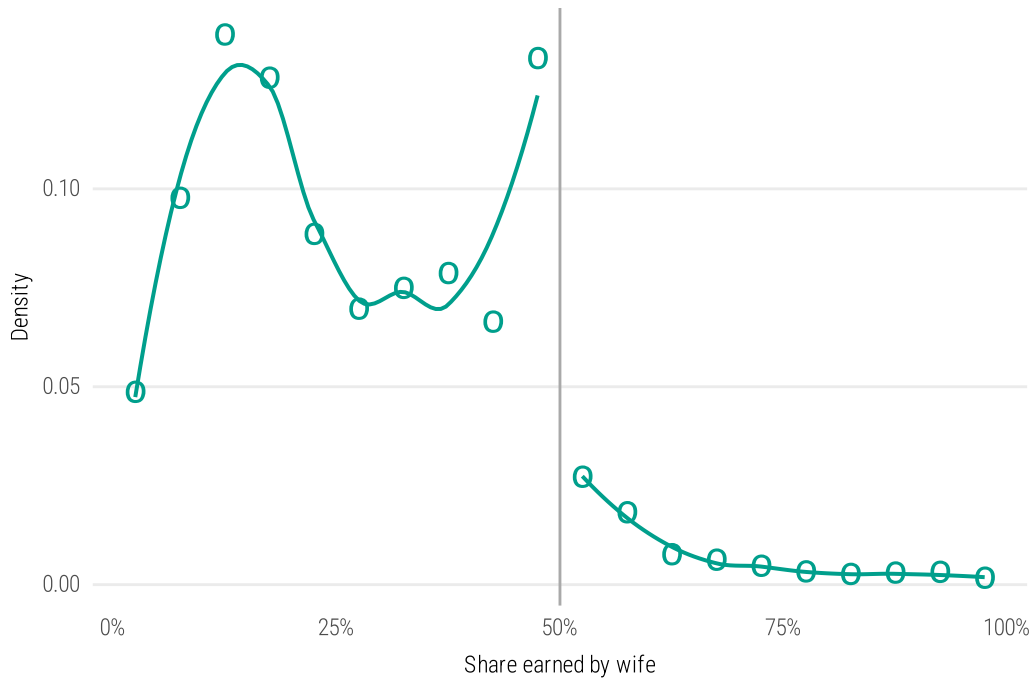


Figure 7: Distribution of Relative Earnings. The data is pooled from JPSED 2017-2020. The sample includes married couples aged 25-59. Dual-earner couples only. Each dot represents the density of couples in a 0.05 relative earnings bin. The vertical line indicates where the wife's share of earnings is 0.5. The dashed line is a lowess smoother applied to the distribution, allowing for a break at 0.5.

If there is a penalty for higher wives' earnings, which reflects social norms, women, upon marriage, will be more likely to choose shorter working hours, non-regular jobs, or exit from the labor market. In Figure 8, I show the event study of labor market outcomes since marriage. The sample and the estimation method are explained in Section A.2. My approach follows the recent literature that studies the impact of children on gender gaps, e.g., Kleven, Landais, and Sogaard (2019), but focuses on marriage rather than the childbearing event.⁸

Since JPSED is an individual survey for workers, I cannot observe individuals who exit from the labor market after marriage. Thus, I focus on individuals who are employed before and after marriage. From the figure, we can see a significant decline in earnings, working hours, and the share of regular workers only for women, and a significant increase in domestic labor hours.

If those who exited the labor market were included, the results would likely be reinforced: labor market exit implies zero earnings, zero working hours, and no regular employment, along with a likely increase in domestic labor hours. Therefore, the observed penalties in earnings, regular employment share, and working hours, as well as the increase in domestic work, would be even more pronounced.

⁸For a set of European countries, Berniell et al. (2022) estimate the separate effects of marriage and childbearing on women's labor market outcomes, and find a relatively small role for marriage. Kleven, Landais, and Leite-Mariante (2025) compute the marriage and child penalty on employment rate for Brazil, China, Japan, Mauritius, Rwanda, Sweden, the United Kingdom, and Zambia. Herold and Wallossek (2023) calculates marriage earning gaps with German administrative data.

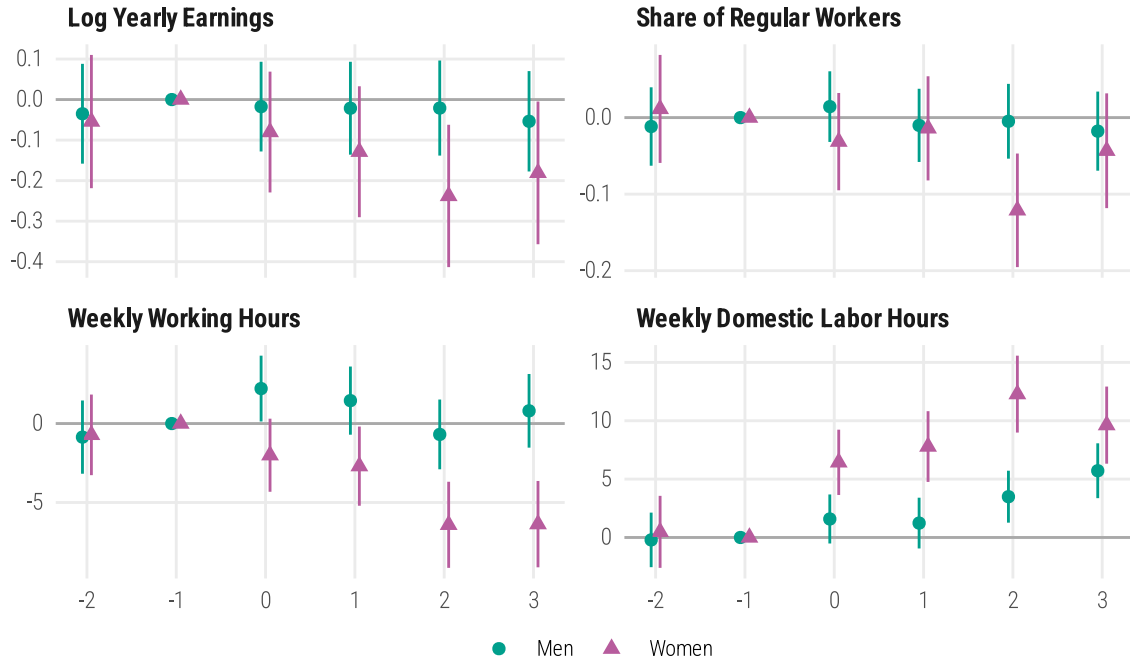


Figure 8: Impact of Marriage on Labor Outcomes. The data is from JPSED 2016-2025. For sample construction details, see Section A.2. Each point represents the estimated coefficient relative to the year prior to marriage ($k = -1$). Error bars indicate 95% confidence intervals based on standard errors clustered at the individual level.

While these social norms are observed in many countries, they are particularly strong in Japan. The World Values Survey includes a question asking, “If a woman earns more money than her husband, it’s almost certain to cause problems.” Respondents answer on a 1-5 scale, where 1 indicates “strongly agree” and 5 indicates “strongly disagree”. Figure 9 plots the mean of the score by country. Among these high-income countries, Japan has the strongest social norms.

**If a woman earns more money than her husband,
it's almost certain to cause problems.**

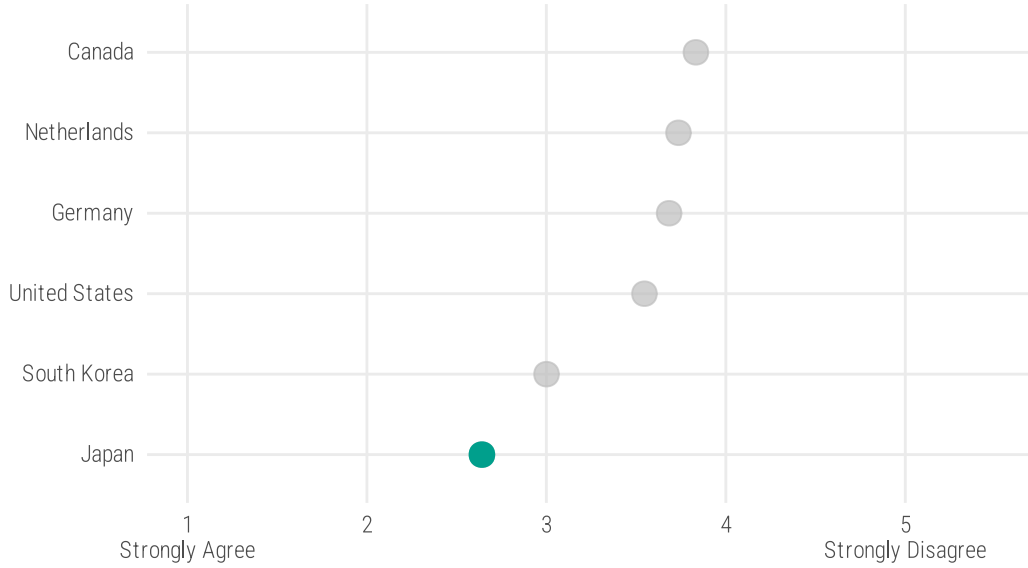


Figure 9: Social Norms on Wives' Earnings. Each data point represents the mean agreement score by country (1: Strongly Agree to 5: Strongly Disagree). Source: World Value Survey Wave 7 (2017-2022).

3 Model

3.1 Overview

The model economy is populated by married couples that consist of a male and a female, denoted by $g \in \{m, f\}$. A couple decides its occupational choices and the allocation of market and domestic labor hours to maximize their joint utility. The occupations can be regular, R , or non-regular, NR . Not working is denoted by NW . If an individual works, they also decide on their market hours, h_m and h_f . Each individual is endowed with one unit of time and a requirement of joint domestic labor hours D . This requirement differs across households, and each household decides their domestic labor hours d_m and d_f , satisfying $D = \zeta(d_m, d_f)$. Each individual is endowed with a productivity (ability) level, denoted by a_g . These ability levels are drawn from a joint distribution for the couple.

3.2 Model Components

Utility Function

$$u(c, h + d) = \log c - \phi \frac{(h + d)^{1+\gamma}}{1 + \gamma},$$

where c is consumption, $h + d$ is the total labor hours (market plus domestic), $\phi > 0$ is a weight on disutility from labor, and $\gamma > 0$ captures the curvature of disutility from labor.

Productivity I also assume that the set of productivities (a_m, a_f) varies across couples. In particular, (a_m, a_f) is drawn from a log-normal distribution.

Convex wage schedules Following Goldin (2014) and Erosa et al. (2022), regular and non-regular jobs in the model differ in how hours worked, h_m , map into effective labor input that determines earnings. For regular jobs, effective labor input is a convex function of hours worked, i.e., the longer an individual works, the higher her effective labor input. This creates incentives to work longer hours since there is an implicit penalty for working short hours. In contrast, the relation between hours worked and effective labor input is linear for non-regular jobs. As a result, if one of the partners cannot supply long hours, they have an incentive to select a non-regular occupation.

Figure 2 shows that 36.1% (46.4%) of regular male (female) workers work exactly 40 hours per week and that 52.3% (26.8%) work more than that. This suggests that 40 hours of work per week is a standard requirement for regular workers and makes it difficult for women to work as regular workers.

To capture these two features, i.e., linear vs. non-linear jobs and a concentration of 40 hours per week, I assume that the production function for regular (non-linear) jobs is given by

$$e(h, a, j = R) = \begin{cases} ah^{1+\theta} & \text{if } h \leq \bar{h} \\ a\bar{h}^\theta h & \text{if } h > \bar{h}. \end{cases}$$

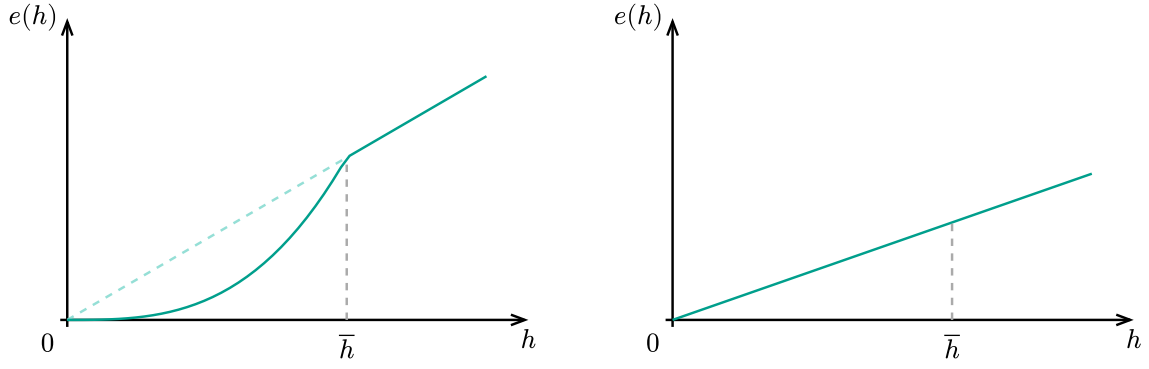
$\bar{h}$ is set to 40 hours per week. The parameter $\theta > 0$ captures the convexity of the earnings curve for regular jobs. The slope of the earnings after 40 hours is the same as that at 40 hours, i.e., $a\bar{h}^\theta$.

For non-regular jobs I assume that the production function is linear:

$$e(h, a, j = NR) = \psi a\bar{h}^\theta h.$$

The parameter $0 < \psi < 1$ represents a wedge that captures the difference in productivity between regular and non-regular jobs. If a regular worker changes to a non-regular job and works the same hours (but more than 40 hours), her earnings are reduced to a fraction ψ of the regular earnings.

The wage schedules for regular and non-regular jobs are illustrated in Figure 10. Since $\theta > 0$, regular workers have an incentive to work longer hours. It also means that if one of the spouses chooses a regular job and works long hours, the other may want to work short hours and select a non-regular job.



(a) Regular Job

(b) Non-Regular Job

Figure 10: Convex Wage Schedule for Regular Jobs vs. Linear Schedule for Non-Regular Jobs.

Domestic Labor Hours Requirement I assume that each household is given a domestic labor hours requirement D and chooses the allocation of d_m and d_f . D is heterogeneous across households, which reflects the fact that the amount of domestic labor hours differs depending on the number of children and their ages. In particular, I assume that D is drawn from a cumulative distribution F_D .

To satisfy the requirement, couples need to allocate their domestic labor hours according to the following joint domestic production function:

$$\zeta(d_m, d_f) := \left(d_m^\xi + d_f^\xi\right)^{\frac{1}{\xi}} = D. \quad (1)$$

If domestic labor hours are perfect substitutes, i.e., $\xi = 1$, the couple will allocate all domestic labor hours to the partner with the lower opportunity cost (wage). However, in reality, couples tend to share domestic labor hours even when there is a large difference in wages between the partners. To capture this feature, I assume that $\xi < 1$, which implies that domestic labor hours are imperfect substitutes.

3.3 Household Problems

Household decisions are made in two steps. First, given productivities (a_m, a_f) and domestic labor hours requirement D , the couple chooses their occupational choices (j_m, j_f) to maximize their expected joint utility. Second, given the occupational choices, they decide the allocation of market and domestic labor hours to maximize their joint utility.

Household Allocations

Given their occupational choices (j_m, j_f) , the couple chooses market hours (h_m, h_f) and domestic labor hours (d_m, d_f) to maximize its joint utility:

$$U^{j_m, j_f} = \max_{c_m, c_f, h_m, h_f, d_m, d_f} u(c_m, h_m + d_m) + u(c_f, h_f + d_f) - \delta \cdot \mathbb{1}\{e_f > e_m\},$$

subject to

$$\begin{aligned} c_m + c_f &= e(h_m, a_m, j_m) + e(h_f, a_f, j_f), \\ D &= \zeta(d_m, d_f). \end{aligned}$$

The last term in the joint utility, which is motivated by Figure 7, represents the utility cost from the wife earning, $e_f = e(h_f, a_f, j_f)$, more than the husband, $e_m = e(h_m, a_m, j_m)$. δ captures the social norm that the husband should earn more than the wife (breadwinner norm).

Occupational Choices

The couple chooses their occupational choices (j_m, j_f) to maximize their expected joint utility:

$$j_m, j_f = \arg \max_{j_m, j_f} U^{j_m, j_f} + \varepsilon^{j_m, j_f},$$

where ε^{j_m, j_f} represents the idiosyncratic preference shock for the occupational choice combination (j_m, j_f) . I assume that ε^{j_m, j_f} is independently and identically distributed according to the type-I extreme value distribution. Hence, the probability that the couple selects occupational choices (j_m, j_f) is given by the multinomial logit formula:

$$\Pr(j_m, j_f) = \frac{\exp\left(\frac{U^{j_m, j_f}}{\eta}\right)}{\sum_{j_m' \in \{R, NR, NW\}} \sum_{j_f' \in \{R, NR, NW\}} \exp\left(\frac{U^{j_m', j_f'}}{\eta}\right)},$$

where η is the scale parameter of the extreme value distribution.

Minimum Working Hours

In the model, I also assume that regular and non-regular jobs have minimum working hours requirements, denoted by $\underline{h}_R$ and $\underline{h}_{NR}$, respectively. If an individual chooses to work as a regular (non-regular) worker, they must work at least $\underline{h}_R$ ($\underline{h}_{NR}$) hours per week. This captures the fact that firms often set minimum working hours requirements for their employees. In particular, I set $\underline{h}_R$ at 20 hours per week and $\underline{h}_{NR}$ at 10 hours per week based on the distribution of working hours in Figure 2.

This condition is also necessary to discipline the model since, without it, an individual with a job (R or NR) working zero hours and an individual not working (NW) would yield the same utility, and the probability of choosing R or NR would be ill-defined.

To sum up, the earning schedule is given by

$$\begin{aligned}
e(h, a, j = R) &= \begin{cases} 0 & \text{if } h < \underline{h}_R \\ ah^{1+\theta} & \text{if } h \leq \bar{h} \\ a\bar{h}^\theta h & \text{if } h > \bar{h} \end{cases} \\
e(h, a, j = NR) &= \begin{cases} 0 & \text{if } h < \underline{h}_{NR} \\ \psi a\bar{h}^\theta h & \text{if } h \geq \underline{h}_{NR} \end{cases} \\
e(h, a, j = NW) &= 0.
\end{aligned}$$

4 Calibration

4.1 Calibration Strategy

I assume that occupation-specific productivity levels (a_m, a_f) are drawn from a multivariate log-normal distribution. I set the mean of log productivity for males, $\mu_m = 0$, as a normalization. To reduce the number of parameters, I assume the variance of the log productivity is the same between males and females.

Based on these assumptions, the productivity levels of a couple are drawn from:

$$\log \begin{pmatrix} \alpha_m \\ \alpha_f \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \cdot & \sigma^2 \end{pmatrix} \right).$$

The home requirements also differ across couples according to

$$D \sim \text{Beta}(\alpha, \beta).$$

This heterogeneity represents differences in domestic labor hours based on the number and age of children. One of the advantages of using the Beta distribution is that it is defined on a finite interval, $(0, 1)$, which can be satisfied by one person or shared by both.

There are two parameters that I set exogenously. Following Erosa et al. (2022), I set the curvature of labor disutility, $\gamma = 3$, so that the Frisch elasticity is fixed at a value of $\frac{1}{3}$. Additionally, following Knowles (2013), I set the intra-household elasticity of substitution for domestic labor, $\xi = \frac{2}{3}$, which corresponds to an elasticity of 3.⁹

Given these functional assumptions, there are 9 parameters to be calibrated:

$$\Pi = \left(\underbrace{\theta, \psi}_{\text{production}}, \underbrace{\eta}_{\text{shock}}, \underbrace{\phi}_{\text{preference}}, \underbrace{\sigma, \rho}_{\text{productivity}}, \underbrace{\alpha, \beta}_{\text{domestic labor}}, \underbrace{\delta}_{\text{social norm}} \right).$$

⁹Knowles (2013) estimates the intra-household elasticity of substitution for domestic labor to be around 3 using the British Household Panel Survey data. He uses a generalized CES production function:

$$\left((1 - \eta_0) d_m^{1-\eta_1} + \eta_0 d_f^{1-\eta_1} \right)^{\frac{1}{1-\eta_1}},$$

and estimates $\eta_0 = 0.475$ and $\eta_1 = 0.33$. η_0 is highly close to 0.5, indicating that husbands and wives equally share domestic labor in terms of efficiency units, which is consistent with the functional form of (1).

The set of parameters Π is estimated to obtain the best possible fit for the model's predictions. Letting D_i and $M_i(\Pi)$ denote the i -th data target and the model's solution for this target, the calibration minimizes the following objective function:

$$\min_{\Pi} \sum_i \left[\frac{D_i - M_i(\Pi)}{D_i} \right]^2. \quad (2)$$

4.2 Moments

Table 1 summarizes the calibrated parameters and their corresponding target moments. While the model parameters are jointly estimated, each is associated with the moment it most directly impacts.

The production parameter θ is determined by the share of regular workers. The relative utility of non-regular jobs influences the share of non-regular workers, which is captured by the idiosyncratic shock parameter η . The wage penalty parameter ψ is pinned down by the log wage differential between regular and non-regular jobs, while the labor disutility ϕ is calibrated to the mean working hours of male regular workers.

The productivity distribution parameters (σ, ρ) are targeted to the gender wage gap for regular workers, the dispersion of log wages, and the spousal correlation in earnings. Parameters governing domestic labor (α, β) are captured by the mean and standard deviation of domestic labor hours. Finally, the social norm parameter δ is calibrated to match the share of households where the wife earns more than the husband.

Importantly, female market outcomes are not targeted, except for the regular wage gap. Thus, the model's predicted gender gaps emerge endogenously from the structural asymmetries generated by the social norm δ .

Table 1: Calibration Results of Baseline Model

Parameter	Value	Target	Data	Model
θ	2.65	Share of regular workers, males	0.90	0.87
η	0.20	Share of non-regular workers, males	0.09	0.09
ψ	0.55	$\overline{\log w_{m,R}} - \overline{\log w_{m,NR}}$	0.64	0.75
ϕ	12.86	$\overline{h_{m,R}}$	0.40	0.40
σ	0.63	$sd(\log w_{m,R})$	0.62	0.63
ρ	0.53	$\text{Corr}(\log e_{m,R}, \log e_{f,R})$	0.21	0.21
α	0.13	$\overline{d_{f,R}}$	0.22	0.22
β	0.55	$sd(d_{f,R})$	0.14	0.14
δ	0.84	Share of $e_f > e_m$	0.07	0.07

Notes: The first and second columns show the estimated values of the model parameters. The third column lists the calibration targets, while the fourth and fifth columns report their moment values in the data and the baseline model, respectively. The parameters are estimated by minimising the distance between the data and model moments defined by (2).

4.3 Estimated Results

The second column in Table 1 shows the estimated parameter values. By comparing the fourth and the fifth columns, which are the moments in the data and the model, we can see that the model matches all the targets reasonably well. The convex wage schedule parameter $\theta = 2.65$ is sufficiently larger than zero, indicating that regular jobs have a non-linear earnings schedule and penalize short working hours. The wedge of non-regular jobs $\psi = 0.55$ implies that non-regular jobs pay approximately 45.3% less than regular jobs for the same hours worked beyond 40 hours per week.¹⁰ The gender correlation in skills $\rho = 0.53 > 0$ implies positive assortative mating. The requirement of home hours has the mean $0.19 \left(= \frac{\alpha}{\alpha + \beta} \right)$, which corresponds to 21.6 hours per week in the case where only one spouse provides domestic labor. The rest of the parameters $\eta = 0.20$, $\sigma = 0.63$, $\psi = 0.55$, $\delta = 0.84$ show reasonable values and contribute to matching the moments.

5 Baseline Economy

5.1 Occupational Choices and Hours Worked

I show how the model economy performs along dimensions that are not targeted in the calibration. In Table 2, I present the occupational choice matrix for husbands and wives. The rows are husbands' jobs and the columns are wives' jobs, and each cell represents the share of couples with that occupational combination. Overall, the model explains the distribution of occupations well. However, the share of non-working husbands is higher in the model than in the data. This

¹⁰One of the reasons why non-regular jobs pay less is that non-regular workers are less likely to have job training. According to JPSED, 42.8% (37.7%) of male (female) regular workers have additional opportunities for training given by their employers (off-the-job training), while 27.1% (21.6%) of male (female) non-regular workers do.

suggests that the inflexibility of regular jobs (convexity of their earnings) is excessive or the social norms on gender roles are insufficient.

Table 2: Occupational Choice in Baseline Model

Husband	Wife		
	Regular	Non-regular	Not-work
Data			
Regular	0.35	0.38	0.17
Non-regular	0.01	0.07	0.01
Not-work	0.01	0.01	0.00
Model			
Regular	0.18	0.32	0.36
Non-regular	0.04	0.03	0.02
Not-work	0.04	0.00	0.00

Notes: The table shows the occupational choice of husbands and wives. The rows represent husbands' occupations, and the columns represent wives' occupations. Each cell reports the share of couples in the corresponding occupational combination.

Finally, I compare the time allocation of couples in each occupation in Table 3 . The model successfully replicates the characteristic patterns: First, husbands work longer than wives. Second, regular workers work longer than non-regular workers.

Table 3: Allocation of Weekly Working Hours of Baseline Model

Husband	Wife	Data		Model	
		Husband	Wife	Husband	Wife
Regular	Regular	44.4	39.7	45.0	34.1
Regular	Non-regular	45.4	23.5	42.5	13.8
Non-regular	Regular	37.0	39.7	23.3	39.6
Non-regular	Non-regular	39.8	25.5	35.7	18.7

Notes: The table shows the allocation of weekly working hours by couple's occupations. Columns 1-2 show the husband's and wife's occupations. Columns 3-4 report the data, and Columns 5-6 report the model predictions.

5.2 Gender Gaps

How do gender gaps in the model economy, which are not directly targeted, compare with the data? Table 4 shows the gender gaps in the aggregate economy. The first column shows the statistics from the data and the second column shows the simulation results. The third column shows the ratio of the model to the data column. The first row, the gender gap in participation, represents the difference in participation rates across genders. The second row, the gender gap in occupation, represents the difference in the share of regular workers. For example, the value

0.53 in the data column represents the difference in the share of regular workers (90.0 % for males and 37.4% for females). The third and fourth rows are the gender gaps in log working hours and wages. One of the most interesting findings is that the model explains almost the entire gender gap in participation rates (205.7%), occupational choices (113.9%), and working hours (156.8%). This is consistent with the social norm against wives' higher earnings. Women have an incentive to reduce their earnings (to zero) by quitting their jobs, changing their occupations, or reducing their working hours. In addition, the model explains a significant proportion of the gender gap in wages (55.6%).

Table 4: Gender Gaps in Baseline Model

	Data	Model	Ratio
Participation	0.16	0.33	206%
Occupation	0.53	0.60	114%
Labor Hours	0.49	0.77	157%
Wage	0.76	0.43	56%

Notes: This table shows gender gap measurements and their values in the data and the model. "Participation" refers to the difference in participation rates between males and females. "Occupation" is the gap in the share of regular workers. "Labor Hours" and "Wage" represent the differences in mean log working hours and mean log hourly wages, respectively. The "Ratio" column reports the ratio of the model gap to the data gap.

6 Regional Variation in Social Norms

In the previous section, I demonstrated that the baseline model, calibrated to the national average, successfully reproduces the key patterns of gender gaps in the Japanese labor market. To further validate the model's mechanism, specifically the role of social norms, I now exploit the regional variation across Japanese prefectures. If the interaction between social norms and labor market structures is indeed the primary driver of gender gaps, then regions with stronger traditional norms should exhibit larger gender disparities, and the model should be able to replicate these cross-regional differences by varying only the social norm parameter.

Previous studies, such as Abe (2013), document substantial cross-prefecture variation in married women's labor force participation that cannot be explained by individual characteristics, suggesting that local factors like social norms play a significant role. This section explicitly measures these local norms and relates them to gender gaps.

6.1 Social Norms Score

To construct a measure of social norms at the prefecture level, I use the "Survey on Awareness of Women's Participation and Advancement in Regional Areas"¹¹ conducted by the Gender Equality

¹¹This is an author translation. The Japanese title is "Chiiki ni okeru Josei no Katsuyaku ni kansuru Ishiki Chōsa".

Bureau, Cabinet Office, Government of Japan in February 2015. The survey includes a battery of questions on attitudes toward traditional gender roles in the household and the workplace.

I choose the following five items from the survey as the basis for constructing the social norms score:

1. *My ideal family is one in which “the husband works outside the home and the wife takes care of the home.”*
2. *In general, I think that “the husband should work outside the home and the wife should take care of the home.”*
3. *Housework and child-rearing should be done by women.*
4. *Caregiving should be done by women.*
5. *When children are small, it is better for the mother not to work outside the home.*

The responses are on a 4-point scale (“Agree”, “Somewhat agree”, “Somewhat disagree”, “Disagree”). For each item, I take the prefecture-level average of the responses and then take the mean across the five items. Higher scores indicate more conservative attitudes toward gender roles. I use this prefecture-level score as the proxy for local social norms in the subsequent analysis.

6.2 Social Norms and Gender Gaps by Prefecture

Figure 11 shows the relationship between the social norms score constructed above and the four gender gaps in labor market outcomes defined in Section 5.2. Each point represents a prefecture, and the solid line is the linear fit. We can see a clear positive relationship in all four panels, indicating that more conservative social norms are associated with larger gender gaps in participation, occupation, hours worked, and wages.



Figure 11: Social Norms and Gender Gaps by Prefecture. The figure shows the relationship between social norms scores (x-axis) and gender gaps in labor market outcomes (y-axis) across prefectures. Social norms score is defined in Section 6.1 and higher score represents more conservative attitudes toward gender roles. The gender gaps measurement is defined in Section 5.2. Each point represents a prefecture, and the solid line is the linear fit (OLS).

Next, Figure 12 presents two additional relationships that further illustrate the role of social norms. The left panel shows the relationship between the social norms score and the share of wives earning more than their husbands, which is one of the targeted moments in the baseline economy. The right panel shows the relationship between the social norms score and the flexibility in working time among female regular workers, which is a key mechanism in the model as discussed in Section 2.3. While we can see a clear negative relationship in the left panel, the right panel shows a weaker relationship. As discussed in Section A.4, the social norm score does not have a significant relationship with most of the flexibility measures.¹² Taken together, these patterns suggest that regional variation in gender gaps is more directly related to social norms. Motivated by this observation, I next show that allowing only social norms to vary across prefectures is sufficient to account for a large share of the observed regional variation in gender gaps.

¹²There is a significant negative relationship for the flexibility of female non-regular workers. This mostly does not affect the results. This is because the flexibility of non-regular jobs is already high in both the model and the data.

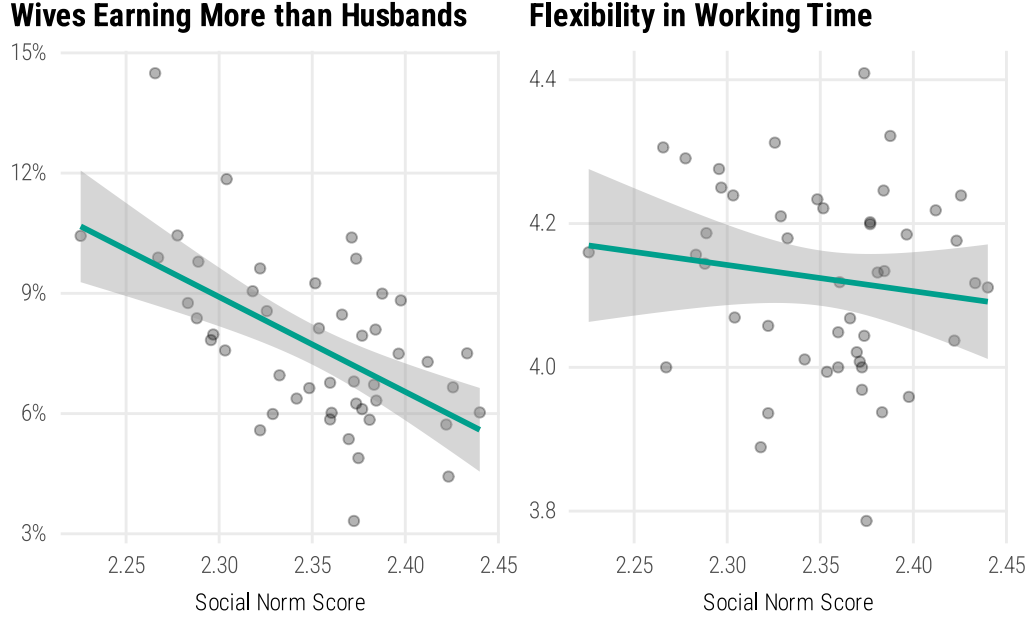


Figure 12: Social Norms, Share of Wives Earning More than Husbands, and Work Flexibility. The left figure shows the relationship between social norms scores (x-axis) and the share of wives earning more than husbands (y-axis) across prefectures. The right figure shows the relationship between social norms scores (x-axis) and the flexibility in working time (y-axis) among female regular workers across prefectures. The flexibility measurement is defined in Section 2.3 and higher value represents more flexible working arrangements.

6.3 Model Prediction

This section assesses whether the baseline model calibrated at the national level can account for a large share of the observed cross-prefecture variation in gender gaps by allowing only the social-norm parameter to vary across prefectures.

Let the four measurements of gender gaps be denoted by a vector $\mathbf{g} = (g_1, g_2, g_3, g_4)$, where each element corresponds to participation, occupation, labor hours, and wages, respectively. Let $\mathbf{g}_p^{\text{data}}$ denote the vector computed from the data for prefecture p . I assume the baseline model can explain the cross-prefecture variation in $\mathbf{g}_p^{\text{data}}$ through prefecture-level differences in the social-norm parameter δ_p , holding the remaining model parameters fixed at the nationally calibrated values $\mathbf{X} = \hat{\mathbf{X}}$:

$$\mathbf{g}_p^{\text{data}} = G(\delta_p; \hat{\mathbf{X}}) + \mathbf{g}_0 + \Xi_p,$$

where $G(\cdot)$ maps model parameters to predicted gender gaps, $\mathbf{g}_0$ is a prefecture-invariant residual component, and Ξ_p is an idiosyncratic error term.

Crucially, I pin down $\mathbf{g}_0$ using the nationally calibrated baseline economy. Let $\mathbf{g}_{\text{JPN}}^{\text{data}}$ be the vector of gender gaps measured in the aggregate (Japan-wide) data, and let $\hat{\delta}$ denote the baseline value of the social-norm parameter in that calibration. I define

$$\mathbf{g}_0 := \mathbf{g}_{\text{JPN}}^{\text{data}} - G(\hat{\delta}; \hat{X}).$$

That is, $\mathbf{g}_0$ is the difference between the gender gaps observed in the data and the gender gaps predicted by the calibrated baseline model economy, and I treat this wedge as common across prefectures when interpreting cross-regional variation. In other words, $\mathbf{g}_0$ and $G(\hat{\delta}; \hat{X})$ correspond to the first and second columns of Table 4.

Similarly, I assume the model predicts the share of wives who earn more than their husbands π_p in prefecture p as a function of δ_p :

$$\pi_p = f(\delta_p; \hat{X}) + \zeta_p,$$

where $f(\cdot)$ maps model parameters to the share of wives earning more than husbands, and ζ_p is an idiosyncratic error term. Since this share is one of the targeted moments in the baseline economy, I can pin down the prefecture-invariant component to zero.

To relate the model-implied share π_p (and hence δ_p) to the observed score s_p , I treat s_p as a noisy proxy that is informative about the same underlying prefecture-level heterogeneity. Because the model delivers π (via $f(\delta; \hat{X})$), the relevant link for plotting predictions against the score is the conditional expectation of the score given the share. I therefore use the best linear predictor

$$\mathbb{E}[s_p \mid \pi_p] \approx \alpha_0 + \alpha_1 \pi_p,$$

estimated by regressing s_p on π_p in the prefecture-level data.

Combining this measurement equation with the model mapping from δ to moments yields a parametric prediction curve indexed by δ :

$$\hat{s}(\delta) = \hat{\alpha}_0 + \hat{\alpha}_1 f(\delta; \hat{X}), \quad \hat{\mathbf{g}}(\delta) = G(\delta; \hat{X}) + \mathbf{g}_0.$$

The dashed line in Figure 13 plots $(\hat{s}(\delta), \hat{\mathbf{g}}(\delta))$ as δ varies over the model's simulated support.



Figure 13: Social Norms and Gender Gaps by Prefecture: Data vs. Model Prediction. Points show prefecture-level gender gaps in the data plotted against the social norm score. The solid line in each panel is the linear fit in the data (OLS). The dashed line is the model-implied prediction curve: for each δ on the model's simulated support, the model generates the share of wives earning more than husbands $f(\delta; \hat{X})$ and the implied gender gaps $G(\delta; \hat{X}) + g_0$. To express the model prediction on the score axis, I map $f(\delta; \hat{X})$ into a predicted score using the best linear predictor from the prefecture-level regression of the score on that share.

As Figure 13 shows, the model-implied prediction curve and the linear fit in the data nearly coincide in each panel. This close overlap implies that cross-prefecture variation in gender gaps can be accounted for largely by variation in the model's social-norm parameter δ , holding all other calibrated parameters fixed at their national values. In this sense, regional heterogeneity in social norms is a sufficient single-index source of variation for organizing the prefecture-level patterns in participation, occupation, hours worked, and wages.

7 Counterfactual Simulations

7.1 Counterfactual Scenarios

What accounts for the gender wage gap and part-time work in Japan? In the model, two factors play a key role in answering these questions: **job inflexibility** and **social norms**. As discussed in Section 2, Japan has relatively strong social norms on wives' earnings, and the job inflexibility of Japanese regular workers is high.

To quantify the impact of these factors, I conduct two counterfactual simulations.

1. Flexible Regular Jobs: I run simulations with flexible regular jobs. In the baseline model, regular jobs have a non-linear wage schedule with a penalty for short hours. In this counterfactual, the earning schedule $e(\cdot)$ is set to be linear:

$$e(a, h, j = R) = \begin{cases} 0 & \text{if } h < \underline{h}_R \\ a\bar{h}^\theta h & \text{if } h \geq \underline{h}_R \end{cases}$$

This removes the pecuniary penalty for working shorter hours in regular jobs, allowing us to isolate the effect of job inflexibility.

2. Outsourcing of Housework: I consider the outsourcing of housework. Recent literature emphasizes that the marketization of home production is a key determinant of female labor supply and the gender wage gap. For instance, Cortés and Pan (2019) show that the availability of market substitutes for household production allows high-skilled women to work longer hours, thereby narrowing the gender wage gap. Similarly, Duval-Hernández, Fang, and Rachel Ngai (2023) find that cross-country differences in the marketization of home production explain a significant portion of the variation in female working hours. Low-skilled immigration often facilitates this marketization by lowering the cost of domestic services (Furtado 2016; Cortés and Tessada 2011).

In contrast, Japan’s usage of external household services remains exceptionally low. According to the Family Income and Expenditure Survey 2021 of the Japanese Statistics Bureau, households of two or more persons spend only 2.6 euros per year.¹³ This is partly due to restrictive immigration policies.

Suppose households could purchase household services in the market in Japan. How would this affect gender gaps? To answer this question, I extend the baseline model as follows.

$$U^{j_m j_f} = \max_{h_m, h_f, d_m, d_f} u(c_m, h_m + d_m) + u(c_f, h_f + d_f) - \delta \cdot \mathbb{1}\{e_f > e_m\},$$

subject to

$$c_m + c_f + pd = e_m(h_m, a_{m,j}, j_m) + e_f(h_f, a_{f,j}, j_f),$$

$$D = \left(d_m^\xi + d_f^\xi + d^\xi \right)^{\frac{1}{\xi}}.$$

where d is purchasable housework hours, and p is its price. This model allows couples to purchase external household services and to satisfy the home hours constraint. In other words, it encourages high-skilled men and women to work more hours in the labor market.

Given the scarcity of housework services, we can consider that the price of external housework, p , in the benchmark economy is prohibitively expensive in Japan, so $d = 0$. Then, I conduct a simulation of the case where the price of housework services is affordable. In particular, I set its price to the mean and median wage of a non-regular job in the benchmark economy ($p = \psi \bar{h}^\theta$). Here, I assume that housework services are provided by non-regular workers. The rest of the parameters are fixed at their calibrated values from the baseline model.

¹³I use the category “540 Housekeeping services”.

7.2 Results

Table 5 compares the choice of occupations between the baseline, the flexible regular job case, and the outsourcing case. In the flexible regular job case, we can see an increase in the share of couples with regular jobs (from 18.5% to 39.1%). This is consistent with the fact that job inflexibility is one of the main reasons for women choosing non-regular jobs (Figure 5).

Similar to the case of flexible regular jobs, the outsourcing case also shows an increase in the share of couples with regular jobs (from 18.5% to 51.6%). This is because outsourcing housework reduces the burden of domestic labor (Table A.1), thereby allowing wives to work more in regular jobs.

Table 5: Comparison in Occupational Choice

Husband	Wife		
	Regular	Non-regular	Not-work
Baseline			
Regular	0.18	0.32	0.36
Non-regular	0.04	0.03	0.02
Not-work	0.04	0.00	0.00
Flexible Regular Jobs			
Regular	0.39	0.24	0.25
Non-regular	0.05	0.02	0.01
Not-work	0.03	0.00	0.00
Outsourcing			
Regular	0.52	0.23	0.16
Non-regular	0.04	0.01	0.00
Not-work	0.03	0.00	0.00

Notes: This table shows the fraction of each combination of couples' occupations. The first panel shows the results of the baseline model. The second panel shows the simulation results of linearizing the wage schedule of regular jobs, fixing other parameters. The third panel shows the results of the counterfactual economy with outsourcing of housework.

Table 6 presents the impact of counterfactuals on gender gaps. The introduction of linear wage schedules substantially reduces gender disparities, eliminating 29.9% of the participation gap, 31.5% of the occupational gap, and 70.1% of the wage gap. However, the labor hours gap remains largely unaffected. Even when the pecuniary penalty for flexibility is removed, the asymmetry in domestic responsibilities persists (Table A.1). As long as social norms compel women to supply the majority of household labor, their opportunity cost of market work remains high, preventing full convergence in hours. Consequently, contrary to the prediction in Goldin (2014), the gender gap does not vanish entirely. The remaining disparities highlight the significant portion of the gender gap that is attributable to social norms rather than structural barriers alone.

Outsourcing eliminates significant proportions of all types of gender gaps (58% in participation, 46% in occupation, 44% in working hours, 69% in wages). Unlike the case of flexible regular jobs, the gender gap in labor hours also narrows significantly. This reduction occurs because outsourcing alleviates the disproportionate burden of domestic work on women, allowing them to increase their market hours.

Table 6: Comparison in Gender Gaps

	Baseline	Flexible Regular Jobs	Outsourcing
Participation	0.33	0.23 (70.1%)	0.14 (41.6%)
Occupation	0.60	0.41 (68.5%)	0.32 (53.8%)
Labor Hours	0.77	0.66 (86.1%)	0.43 (56.3%)
Wage	0.43	0.13 (29.9%)	0.13 (31.0%)

Notes: This table shows gender gaps measurements for baseline, flexible regular job cases, and outsourcing cases. The brackets show the percentage of gaps remaining compared to the baseline.

Finally, I examine how the effect of these counterfactual scenarios interacts with social norms. Figure 14 illustrates the heterogeneous effects of the counterfactual scenarios across different levels of social norm intensity (δ). In both the flexible regular job and outsourcing cases, the reduction in gender gaps is larger when δ is higher. This result implies that structural reforms such as promoting labor market flexibility or subsidizing household services are particularly effective in societies with strong gender norms, such as Japan.

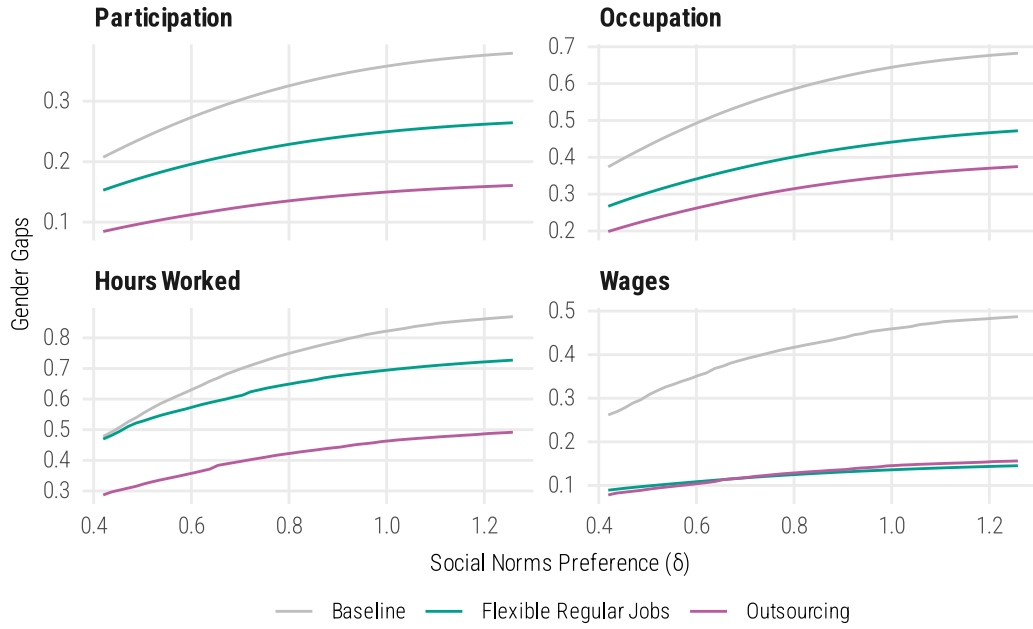


Figure 14: Interaction between Social Norms and Counterfactual Scenarios. The figure shows the relationship between social norms preference (δ) and gender gaps under the baseline, flexible regular job, and outsourcing scenarios.

8 Conclusion

This paper investigates the interaction between job inflexibility and social norms in generating gender gaps. I develop a quantitative model of household labor supply that captures the trade-off between inflexible, high-paying “regular” jobs and flexible, low-paying “non-regular” jobs, alongside gender identity norms. The model successfully reproduces key features of the Japanese labor market, demonstrating that the combination of inflexible employment structures and social norms drives women into non-regular employment. Furthermore, I validate this mechanism by exploiting regional variation across Japanese prefectures, showing that the model can account for cross-regional differences in gender gaps solely through differences in social norms.

Counterfactual simulations show that increasing job flexibility significantly reduces wage and occupational gaps. However, the gender gap in working hours persists due to the unequal burden of domestic work. Policies such as affordable household services are necessary to close this gap. Furthermore, the model predicts that the effectiveness of these policies in reducing gender gaps is larger when social norms are stronger.

Appendix

A Supplementary Analysis

A.1 Specifications of Discontinuity of Relative Earnings

Figure 7 indicates the gap in the density of share of earnings between husbands and wives. While this graph simply shows the discontinuous behavior of couples in terms of earnings, the 5% binning may be too coarse. I thus conduct the same analysis with 1% binning. In Figure A.1, we can see discontinuity at 50% of the share of earnings in both specifications. We can also see a mass point at 50% in the 1% binning specification, and the density at exactly 50% is 0.072. The existence of a mass point at 50% is also consistent with the model settings. The utility cost is imposed when a wife earns strictly more than her husband, and they have an incentive to earn the same amount. See Kuehnle, Oberfichtner, and Ostermann (2021) for a more detailed discussion about the mass point at 50% of the share of earnings and the specification of the discontinuity.

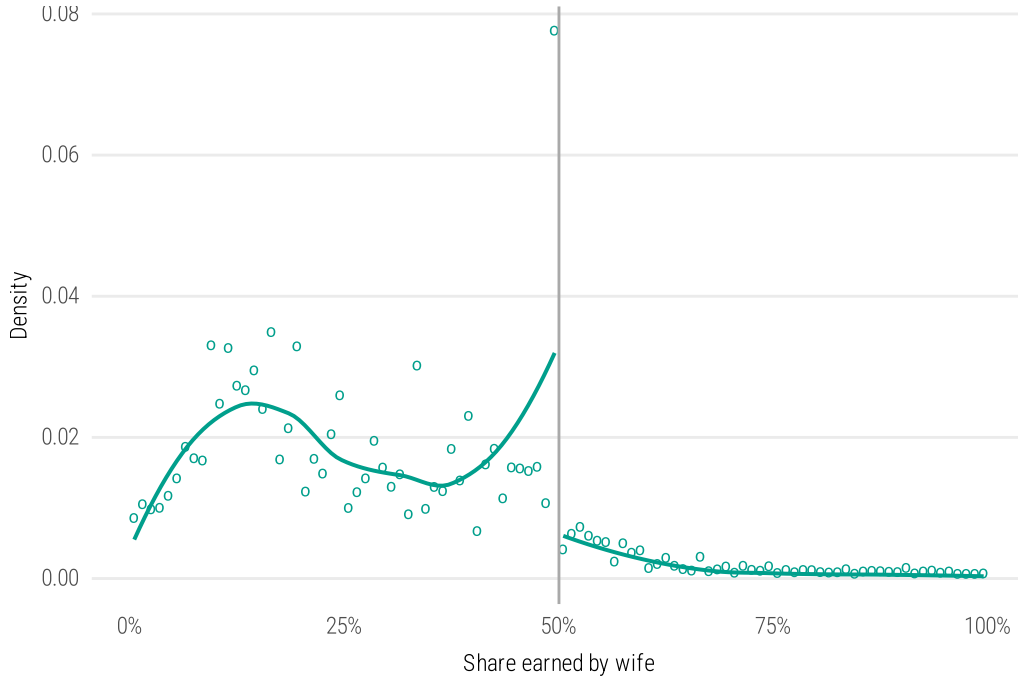


Figure A.1: Distribution of Relative Earnings. The samples are the same as Figure 7. The figure shows the density of the share of earnings between husbands and wives with 1% binning.

A.2 Specification of Marriage Penalty

Figure 8 shows the event study of the following specification:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1, -\infty} \beta_k 1\{e_i + k = t\} + \varepsilon_{it},$$

where α_i is the individual fixed effect, λ_t is the year fixed effect, and e_i is the year of marriage for individual i . The outcome variable y_{it} is log of yearly earnings, share of regular workers, weekly hours worked, and weekly domestic labor hours.

The sample includes people aged 25 to 59 who are observed in all years between 2016 and 2023. I use those who got married between 2019 and 2021 as the treatment group to observe the variables $k = -2, \dots, 3$. As a comparison group, I use those who never got married during the sample period.

A.3 Regional Variation in Social Norms

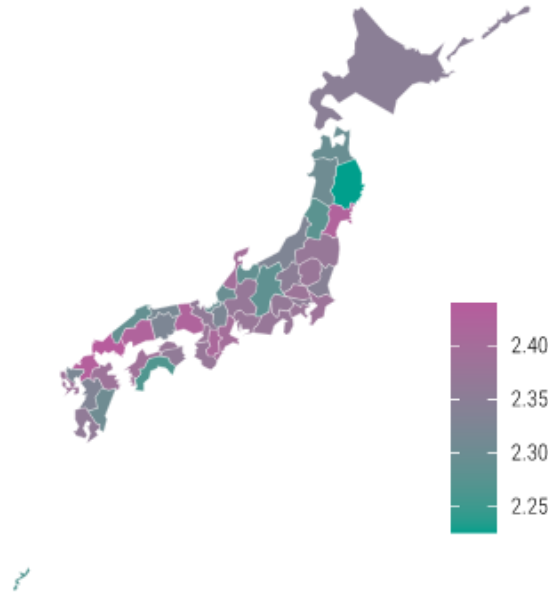


Figure A.2: Regional Variation in Social Norms. The figure shows the social norm score by prefecture in Japan.

A.4 Relationship between Flexibility and Social Norms

Figure A.3 shows the regression coefficients of the flexibility measures on the social norms. The dependent variables are the flexibility measures (day, time, and place) aggregated by prefecture, gender, and employment type. The independent variable is the social norm score in each prefecture. The error bars indicate the 95% confidence intervals with robust standard errors.

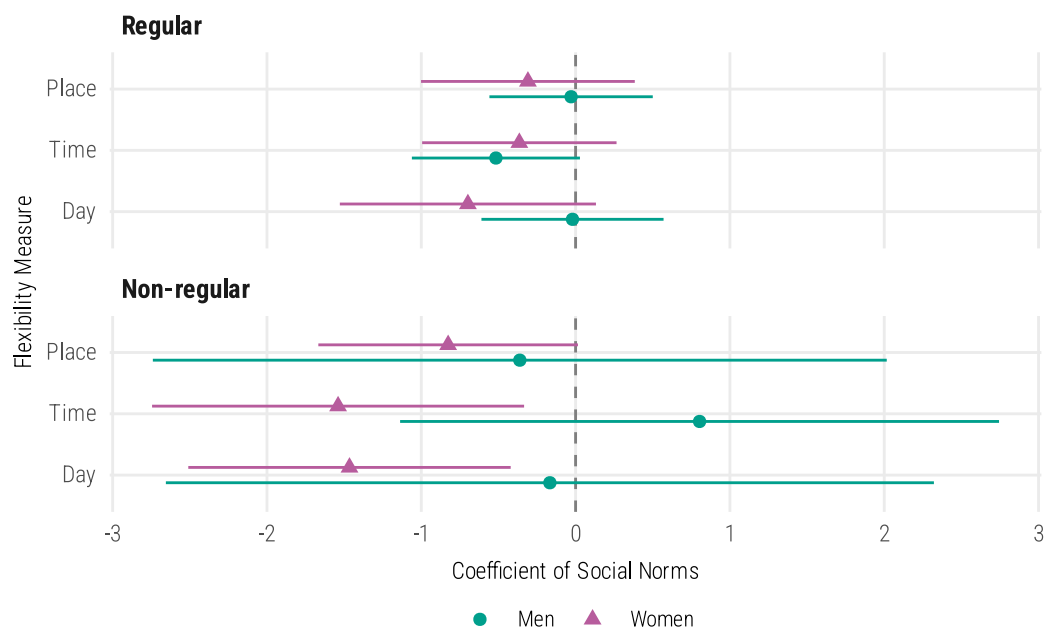


Figure A.3: Relationship between Flexibility and Social Norms. The figure shows the regression coefficients of the flexibility measures on the social norms. The error bars indicate the 95% confidence intervals with robust standard errors.

A.5 Time Allocation in Counterfactuals

Table A.1: Working and Home Hours with Outsourcing d

Husband	Wife	Working Hours		Domestic Labor	
		Husband	Wife	Husband	Wife
Baseline					
Regular	Regular	45.0	34.1	18.9	25.7
Regular	Non-regular	42.5	13.8	22.3	45.2
Non-regular	Regular	23.3	39.6	37.4	23.4
Non-regular	Non-regular	35.7	18.7	26.7	40.1
Flexible Regular Jobs					
Regular	Regular	37.7	24.0	25.3	35.2
Regular	Non-regular	40.1	15.5	23.1	42.3
Non-regular	Regular	30.6	30.4	31.1	30.3
Non-regular	Non-regular	37.1	19.3	25.0	38.8
Outsourcing					
Regular	Regular	55.0	41.2	4.9	11.0
Regular	Non-regular	54.3	25.2	5.7	24.0
Non-regular	Regular	32.1	49.7	19.4	7.3
Non-regular	Non-regular	48.8	29.6	10.8	22.7

Notes: The table compares the allocation of weekly working hours (column 1-2) and domestic labor hours (column 3-4) among the baseline model, the counterfactual economy with flexible regular jobs, and the counterfactual economy with outsourcing.

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